

Marathi ASR Fine-tuning - Summary

Fine-tune a speech-to-text model on Marathi, end-to-end, with a working evaluated pipeline.

Task

Pick one of MT / ASR / TTS and fine-tune it on Marathi. ASR was chosen.

Model

ai4bharat/indicconformer_stt_mr_hybrid_ctc_rnn_large - Conformer-Large encoder (17 layers, 512-dim, 115M params) with a hybrid CTC + RNNT decoder (129M params total). Already Marathi-capable; the base checkpoint's own weights and 22-language shared tokenizer were restored and fine-tuned as-is, not retrained from scratch.

Key decisions

- **Dataset:** used OpenSLR SLR64 (Marathi speech corpus, CC BY-SA 4.0) for fine-tuning.
- **Framework and code structure:** built on NVIDIA NeMo (PyTorch Lightning as the underlying training engine) for the model, data pipeline, and training loop. Code is split into independent, single-purpose scripts - environment setup, data preparation, training, and evaluation/packaging - so each stage can be run, tested, and re-run on its own.
- **Fine-tuning strategy:** small learning rate (2e-5), all parameters trainable, to adapt without discarding pretrained acoustic knowledge.
- **Ablation:** a second run trained only the CTC/RNNT decoder heads with the Conformer encoder frozen, to compare against full fine-tuning.

Results

Word Error Rate (WER) on a held-out test split, lower is better:

Model	Test WER	Epochs
Base checkpoint (no fine-tuning)	0.1540	-
Full fine-tune (all params)	0.1009 (best)	1
Heads-only fine-tune (encoder frozen)	0.1215	1

Full fine-tuning improves ~5.3 points over the base model; heads-only improves ~3.25 points but confirms the encoder's larger capacity matters. The best checkpoint (full fine-tune) is the one shared as the deliverable.

Limitations

- **Catastrophic forgetting risk:** full fine-tuning updates all 129M parameters, including the pretrained encoder, on a small (~3h) fine-tuning set. This can degrade the model's broader, general acoustic knowledge as it adapts to the new data - a risk the heads-only run (frozen encoder) partially trades off against, at the cost of a smaller WER improvement.
- **Limited data and training:** the fine-tuning set is small and trained for very few epochs, so results are directional evidence of a working pipeline, not a fully converged model.
- **Single-speaker-gender source data:** SLR64 has female-only speakers, so the WER improvement is somewhat speaker/gender-skewed rather than broadly representative.

Reproducibility

Clone the repo, run `bash setup.sh`, extract the checkpoint, then:

```
python scripts/infer_eval.py --finetuned-model checkpoint.nemo --manifest
data/processed/test_manifest.jsonl
```